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Increasing mean arterial pressure during cardiac surgery does not reduce the rate of postoperative acute kidney injury

A Azau,[1] P Markowicz,[1] JJ Corbeau,[1] C Cottineau,[1] X Moreau,[1] C Baufreton[2] and L Beydon[1]

Abstract

Introduction: We hypothesized that the optimization of renal haemodynamics by maintaining a high level of mean arterial blood pressure (MAP) during cardiopulmonary bypass (CPB) could reduce the rate of acute kidney injury (AKI) in high-risk patients.
Methods: In this randomized, controlled study, we enrolled 300 patients scheduled for elective cardiac surgery under cardiopulmonary bypass. All had known risk factors of AKI: serum creatinine clearance between 30 and 60 ml/min for 1.73m² or two factors among the following: age >60 years, diabetes mellitus, diffuse atherosclerosis. After a standardized fluid loading, the MAP was maintained between 75-85 mmHg during CPB with norepinephrine (High Pressure, n=147) versus 50-60 mmHg in the Control (n=145). AKI was defined by a 30% increased of serum creatinine (sCr). We further tested others definitions for AKI: RIFLE classification, 50% rise of sCr and the need for haemodialysis.
Results: The pressure endpoints were achieved in both the High Pressure (79 ± 6 mmHg) and the Control groups (60 ± 6 mmHg; p<0.001). The rate of AKI did not differ by group (17% vs. 17%; p=1), whatever the criteria used for AKI. The length of stay in hospital (9.5 days [7.9-11.2] vs. 8.2 [7.1-9.4]) and the rate of death at day 28 (2.1% vs. 3.4%) and at six months (3.4% vs. 4.8%) did not differ between the groups.
Conclusion: Maintaining a high level of MAP (on average) during normothermic CPB does not reduce the risk of postoperative AKI. It does not alter the length of hospital stay or the mortality rate.

Keywords

cardiac surgery; acute kidney injury; arterial pressure; cardiopulmonary bypass; primary prevention

Introduction

Postoperative acute kidney injury (AKI) is a frequent complication of cardiac surgery with cardiopulmonary bypass (CPB).[1] Its rate depends on the criteria used to assess the diagnosis.[2] However, AKI remains an independent risk factor for postoperative mortality.[3,4]

Many factors related to surgery have been identified, e.g. inflammation elicited by CPB, nephrotoxins, ischaemia-reperfusion, oxidative stress, metabolic factors, embolisms, neurohormonal activation and comorbidities.[5] Avoidance of nephrotoxic agents (radiocontrast agents, aminoglicosides, aprotinin…), as well as perioperative optimization of general and renal haemodynamics represent the only factors clinicians can work out to preserve renal function.[6] Moreover, hypotension during CPB is known to favour postoperative AKI.[7] Yet, increasing haemodynamics during CPB is easily achievable by thorough haemodynamic monitoring and goal-oriented administration of fluids and vasoactive agents. Worthy of note, noradrenaline has been shown as a renal vasodilator in conscious dogs.[8] Such an approach has been successfully validated in septic shock[9] and postoperatively to major non-cardiac surgery.[10] Further, Gold

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et al.,[11] in a randomized study, tested the advantage of maintaining the MAP between 80 and 100 mmHg in comparison to 50-60 mmHg during hypothermic CPB for coronary grafting. They found that the overall incidence of combined cardiac and neurological complications was significantly lower in the high MAP group. Unfortunately, they did not study renal function. These studies are contradictory to the findings by Urzua et al.[12] and Sirvinskas et al.[13,14] who showed that raising the MAP did not change renal function postoperatively.

Moreover, the clinical profile of patients referred to cardiac surgery centres has changed over time[15,16] and today's patients disclose more risk factors for postoperative AKI than in the past.[17] In addition, patients more frequently undergo valve or combined surgery, which have been identified as major independent risk factors for AKI.[13]

The goal of this study was to re-assess whether maintaining high arterial pressure (>80 mmHg) during normothermic CPB after an adequate fluid loading would reduce postoperative AKI in patients selected to be at high-risk for renal dysfunction and being representative of present-day patients undergoing cardiac surgery.

Methods

Study design

This single-centre, prospective, randomized, controlled study had been approved by our ethical committee in June 2007. It was conducted from January 2008 to June 2010 as a single, blinded trial in our university hospital. All patients had given written informed consent before inclusion.

Patients

All consecutive patients with the following inclusion criteria were enrolled: elective cardiac surgery, including coronary artery bypass graft (CABG), valvular surgery or reconstructive surgery of the ascending aorta performed under normothermic CPB in patients with known risk factors for acute kidney injury. These risk factors were: a serum creatinine clearance between 30 and 60 ml/min for 1.73m^2 or two factors among the following: age >60 years, diabetes mellitus, diffuse atherosclerosis. Exclusion criteria were: infusion of a radiocontrast agent or treatment by a nephrotoxic agent one and three weeks before surgery, respectively; chemotherapy within the three last months; liver cirrhosis; heart failure (left ventricular ejection fraction <30%); renal artery stenosis; pulmonary hypertension (systolic pulmonary pressure >60 mmHg); endocarditis; surgery requiring hypothermic CPB. Patients who eventually disclosed a major perioperative complication (shock, emergent re-operation), identified as AKI cause, were excluded subsequently. Randomisation was performed using opaque sealed envelopes.

Surgery

General anaesthesia was induced with propofol, sufentanil and pancuronium. Maintenance was achieved with isoflurane before and after CPB and with propofol during CPB. Following anaesthesia induction, a systematic 12 ml/kg saline infusion load was administered in all patients. It was renewed whenever deemed necessary by the anaesthetist in charge of the case, according to the haemodynamic parameters.

Haemodynamics were monitored via a SvO_2 Swan Ganz catheter (VIP Oxymetry thermodilution catheter,™ Edwards Lifesciences®, Guyancourt, France) and a radial artery catheter (Leader Cath™, Vygon®, Ecouen, France). Tranexamic acid (50 mg/kg) was infused during the pre-CPB period. The CPB has been described earlier[18] and was performed with a roller pump (HL20™ roller pump, Maquet SA®, Ardon, France) with a heparin-coated circuit, including a soft-shell reservoir (closed circuit) (Bioline Jostra-Maquet, Maquet SA France or Carmeda™ Medtronic, Medtronic®, Paris, France). CPB flow was set at 2.4 L/min/m^2 [19] and further adjusted to maintain the SvO_2 above 70%. The heparin requirement was assessed by a dose-response assay, using the heparin management system (HMS) Hepcon® (Medtronic, Minneapolis, MN) titrated to achieve and maintain an activated clotting time (ACT) >250 sec for CABG or >350 sec for open heart surgery. No cardiotomy suction was used and pericardial blood loss was processed by a cell saver (Cell Saver5+™, Haemonetics®, Plaisir, France) and re-transfused. The haemoglobin was maintained above 7 g/dL during CPB and above 9 g/dL postoperatively. Cold blood cardioplegia was a mixture of 80% blood and 20% cardioplegia solution with 20 mmol/L potassium (St Thomas' Solution, SLF 103, Aguettant, Lyon, France). The protamine reversal dose was determined after titration by the HMS Hepcon®.

MAP and SvO_2 were recorded every 5 seconds via the cardioscope (IntelliVue MP 80®, Philips Healthcare™, Suresnes, France) with the Philips IIC datacollect (B.00.07) and, subsequently, transferred to a computer for analysis of means; from induction to the start of CPB; from the start to the end of CPB; and from the end of CPB to the end of the intervention. With a Jocap XL (Maquet), arterial pump flow was transferred from the HL20, and the SvO_2, temperature and haemoglobin from the CDI™ Blood Parameter Monitoring System 500 Terumo®, to a computer every 10 seconds for analysis of means from start of full flow to the end.

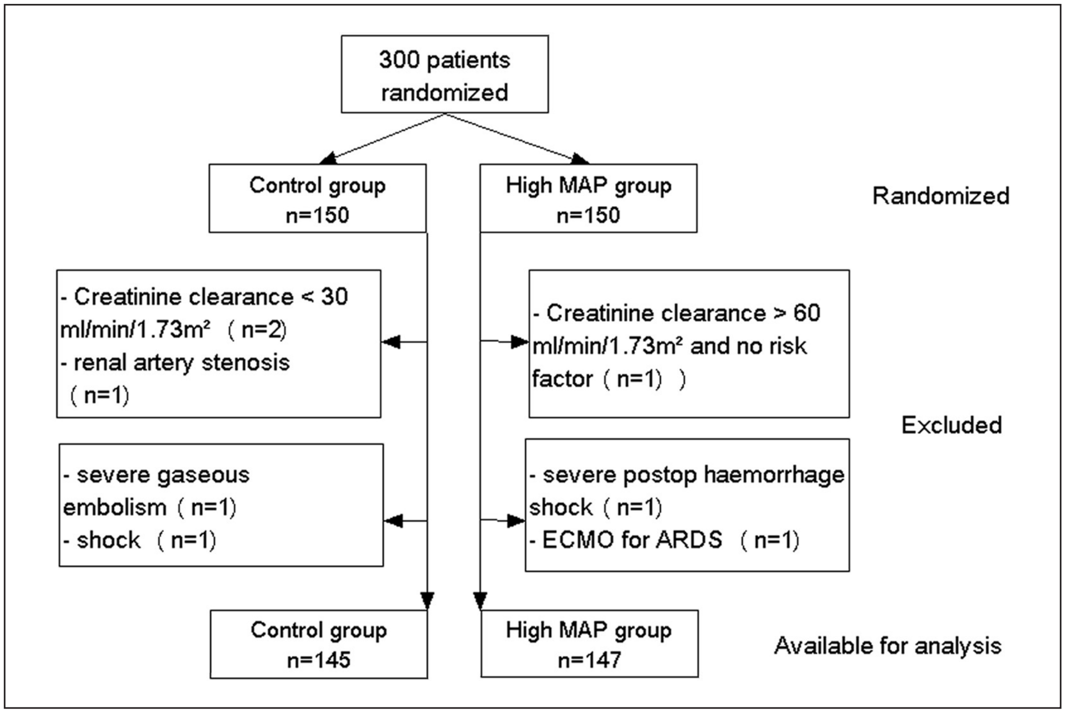

Figure 1. Exclusion figures for Control and High MAP groups.

Intraoperative interventions

In the Control group, the MAP endpoint was 50-60 mm Hg. This level matched conventional clinical CPB routine and vasopressors were administered when the MAP fell below 50 mmHg. In the High Pressure group, the MAP was targeted between 75 and 85 mmHg by infusing norepinephrine. Before and after CPB, and in both groups, the MAP endpoint was 70-90 mmHg, with a SvO_2 >70% until the end of surgery.

Postoperative period

A goal-oriented control of haemodynamics was achieved by combining fluid loading (including a 24 ml/kg basal daily infusion of a 5% glucose solution with NaCl 4 g/L and KCl 2 g/L) with vasoactive agents according to monitoring parameters in order to achieve: a diuresis >0.7 ml/kg/h, a MAP >70 mmHg and a SvO_2 >70%.

The renal resistivity index [RRI = (peak systolic velocity–end-diastolic velocity)/peak systolic velocity)] was measured by trained sonographers, unaware of the group allocation, using a 5 MHz pulsed wave Doppler probe (VIVID7™, GE Healthcare, Chalfont St Giles, UK), as previously described.[20] Briefly, velocity in interlobar or cortico-medullar arteries was assessed three to five times in each kidney and averaged.

Endpoints, analyses and statistics

Serum creatinine (sCr) was serially measured preoperatively (the day before surgery), in the morning of the first five postoperative days and at six months. The maximum increase in sCr during the first five postoperative days was expressed as a percentage of the preoperative value.

A 30% rise in sCr was chosen as the primary endpoint and surrogate for AKI.[21,22] Then, the renal function was assessed at day 28 and at six months after surgery.

As several definitions of acute renal injury have been used in the literature, we further tested four additional surrogate endpoints for AKI: a) the RIFLE “risk” category (50% rise in sCr or glomerular filtration rate decrease >25%, urine output <0.5 ml/kg/h x 6h), b) the RIFLE “injury” category (100% rise in sCr or glomerular filtration rate decrease >50%, urine output <0.5 ml/kg/h x 12h), c) a >50% postoperative rise of sCr and d) the need for haemodialysis.[23,24]

We sequentially used these four criteria in the statistical analysis to explore a possible threshold effect, depending on the criteria used to diagnose AKI.

Neurological complications of surgery (stroke, seizure, transient mental confusion/agitation) were also recorded.

Death date was identified either by medical records received from doctors who cared for the patients after

Table 1. Patient characteristics and preoperative data.

	Control (n=145)	High-MAP (n=147)	p value
Demographic and cardiovascular characteristics			
Gender (Male/Female)	106/39	94/53	0.10
Age (Years)	76 ± 7	76 ± 8	0.77
Weight (kg)	76 ±14	74 ± 13	0.16
Height (cm)	165 ± 9	163 ± 9	0.06
Body surface area (m^2)	1.83 ± 0.2	1.78 ± 0.2	0.09
EuroSCORE (logistic)	7.5 ± 5.5	6.9 ± 4.8	0.31
Left ventricular ejection fraction (%)	59 ± 10	59 ± 11	0.68
Systolic pulmonary arterial pressure (mmHg)	33 ± 10	34 ± 10	0.37
Risk factors for AKI: n [%]			
Preoperative mild chronic insufficiency*	73 [50]	84 [57]	0.29
Diabetes	69 [47.6]	72 [49]	0.82
Age >60 years	143 [99]	145 [99]	1.00
Diffuse atherosclerosis	47 [32]	40 [27]	0.37
Preoperative medications : n [%]			
ACE Inhibitors	70 [48]	76 [52]	0.64
ARB antagonists	47 [32]	47 [32]	1.00
Statin	93 [64]	106 [72]	0.17
Oral hypoglycemic agent	59 [41]	57 [39]	0.81
Insulin	20 [14]	24 [16]	0.62
Beta blocker	78 [54]	91 [62]	0.19
Diuretic	73 [50]	84 [57]	0.29

AKI: acute kidney injury; ACE: angiotensin converting enzyme; ARB: angiotensin receptor blocker.
*a serum creatinine clearance between 30 and 60 ml/min/1.73m².

discharge from hospital or, in the absence of such records, we asked the national registry of deaths.

The number of patients to be included was computed according to our prospective department database, which includes more than 7000 patients. We anticipated a 20% decrease in postoperative renal injury for patients in the high MAP group, with a threshold for significance and a power level of 5% and 80%, respectively. Accordingly, 150 patients in each group were required to show a significant difference.

Statistical analysis was conducted according to an intention-to-treat scheme. Variables expressed in percentage or in counts were compared using a Chi^2 or Fisher exact test. Continuous variables were expressed by mean and standard deviation and compared using analysis of variance. The time for the first dialysis or death were compared between groups using the Kaplan Meier method and Log-rank test. Statistical computations were performed using a bilateral hypothesis with a 0.05 threshold for significance. Analysis was performed with SPSS® (v15. Chicago, IL, USA).

Results

Among the 300 patients enrolled, eight were excluded afterwards as they appeared to meet exclusion criteria (Figure 1). Accordingly, 147 patients in the High Pressure group and 145 patients in the Control group were available for analysis.

Preoperative patient characteristics have been summarized in Table 1. During CPB, the pressure endpoints were achieved in the High Pressure and Control groups (79±6 mmHg *vs.* 60±6 mmHg ($p<0.001$)) (Table 2). From the end of CPB to the end of surgery, the MAP in each group was significantly different, but this difference was marginal from a clinical point of view (70±6 mmHg *vs.* 67±6 mmHg ($p>0.5$) for High Pressure and Control groups, respectively). The MAP was not different by group in the Intensive Care Unit (ICU). Perioperative and renal function data have been summarized in Tables 3 and 4. There was no difference between the two groups. In particular, the type of surgery performed, the usual patient therapies, the risk factors for AKI and preoperative renal function were similar. Eight patients (2.7%) died during the first postoperative month, five in the High Pressure group (3.4%) and three in the Control group (2.1%) (p=not significant).

The rate of AKI, whatever the criteria and thresholds considered, did not differ between the groups (Table 4). Hence, maintaining High Pressure during CPB did not reduce the risk for AKI compared to the Control group.

Table 2. Perioperative haemodynamic parameters.

	Control (n= 145)	High Pressure (n= 147)	P
Pre-CPB period (from the start of anaesthesia to CPB)			
MAP, (mmHg)	73 ± 6	74 ± 6	0.50
MAP goal achieved in each group, n [%]	113 [78]	119 [81]	0.56
SvO_2, (%)	73 ± 6	72 ± 6	0.40
SvO_2 goal achieved in each group, n (%)	104 [75]	96 (71]	0.50
CPB period			
MAP, (mmHg)	60 ± 6	79 ± 6	<0.0001
Patients with a MAP >75 mm Hg, n [%]	–	120 [82]	
SvO_2, (%)	74 ± 5	75 ± 4	0.20
SvO_2 goal achieved in each group, n [%]	124 [86]	126 [87]	0.86
Mean CPB output, (l/min)	4.7 ± 0.5	4.6 ± 0.4	0.38
Number of patients receiving norepinephrine	87	135	<0.0001
Number of patients receiving neosinephrine	10	19	0.12
Cumulative dose of norepinephrine, (mg)	0.52 ± 0.86	2.10 ± 2.17	<0.0001
Cumulative dose of neosinephrine, (mg)	0.21 ± 0.88	0.34 ± 1.18	0.27
Post-CPB period (from the end of CPB to exit from OR)			
MAP (mmHg)	67 ± 6	70 ± 6	0.001
MAP goal achieved in each group, n [%]	48 [33]	77 [53]	0.001
SvO_2, (%)	68 ± 7	67 ± 8	0.10
SvO_2 goal achieved in each group, n [%]	64 [46]	53 [39]	0.31
Fluid loading, (ml)	294 ± 328	272 ± 332	0.6
Norepinephrine: maximum infusion rate, (µg/kg/min)	0.21 ± 0.23	0.37 ± 0.40	<0.0001
Dobutamine: maximum infusion rate, (µg/kg/min)	4.3 ± 3.3	5.3 ± 3.4	0.7
In ICU			
MAP goal achieved in each group, n [%]	94 [65]	83 [56]	0.15
SvO_2 goal achieved in each group, n [%]	55 [38]	52 [37]	0.9
Fluid loading, (ml)	854 ± 689	990 ± 682	0.09
Duration of norepinephrine infusion, (h)	10 ± 11	13 ± 13	0.06
Number of patients receiving norepinephrine, n [%]	107 [75]	130 [90]	0.001
Duration of dobutamine infusion, (h)	17 ± 10	14 ± 4	0.4
Number of patients receiving dobutamine, n [%]	12 [8]	5 [3]	0.85

CPB: cardiopulmonary bypass; MAP: mean arterial pressure; OR: operating room.

The rate of clinically assessed neurological complications did not differ by group. (Pearson Chi-squared test, p=0.67)

The length of hospital stay did not differ between the High Pressure and Control groups (9.5 days [7.9-11.2] *vs.* 8.2 days [7.1-9.4], respectively), nor did the rate of death at six months (5 patients (3.4%) *vs.* 7 patients (4.8%), respectively).

Discussion

Our study showed that a targeted high pressure strategy during CPB and after adequate fluid loading did not reduce postoperative AKI in a population of cardiac surgical patients known to be at increased risk of renal dysfunction. The high pressure endpoint did not reduce the length of stay nor the death rate at day 28 or at 6 months postoperatively.

The high pressure was achieved using a four-fold amount of vasopressors compared to the control group. Noteworthy, all patients received an initial 12 ml/kg fluid loading to correct for preoperative hypovolaemia. Indeed, patients for elective cardiac surgery are often treated with diuretics and fluid restrictive therapy as revealed in our series by the high preoperative blood proteins level. Hence, our haemodynamic goal of high pressure during CPB was fulfilled in 82% of patients in the High Pressure group. Accordingly, the reader may conclude that we did not reach our goal of a target MAP set at >75 mmHg, but this figure corresponds to the mean of all the MAP values recorded every 5 seconds during the whole procedure. A transient decrease in MAP occurred in some instances (e.g. at the onset and separation from CPB, when the surgeon tilted the heart or if cannulae were transiently angled, etc). All these accumulated dips were averaged as we purposely chose

Table 3. Perioperative data.

	Control (n=145)	High-MAP (n=147)	p value
*Type of surgery: n [%]**			
Aortic valve replacement	86 [59]	89 [60]	0.9
CABG	44 [30]	44 [30]	1,00
Mitral valve replacement	4 [3]	11 [7]	0.11
Mitral valve repair	8 [5]	9 [6]	1
Tricuspid valve repair	2 [1]	0 [0]	0.25
Ascending aorta surgery	11 [8]	5 [3]	0.13
Operative parameters			
Baseline haematocrit (%)	41 ± 4	40 ± 4	0.07
Baseline blood proteins (g/l)	75.5 ± 5.3	75.1 ± 5.2	0.6
Fluid loading according to protocol (ml)	1202 ± 456	1221 ± 464	0.72
CPB temperature (°C)	36.7 ± 0.4	36.8 ± 0.4	0.44
CPB duration (min)	113 ± 51	118 ± 43	0.35
Patients requiring per CPB haemofiltration, n [%]	30 [21]	29 [20]	0.88
Quantity of blood processed by the cell saver (ml)	1127 ± 1098	1218 ± 958	0.45
Number of patients transfused [%]	30 [21]	42 [29]	0.14
Number of red blood cells units transfused	0.5 ± 1	0.6 ± 1.1	0.21
Postoperative period			
Peak of troponin (μg/l)	0.69 ± 1.18	0.75 ± 0.98	0.6
Peak of lactates (mmol/L)	1.7 ± 1.1	1.7 ± 1.2	0.5
Postoperative haematocrit at ICU admission (%)	35 ± 4.2	36 ± 4.1	0.005
Postoperative blood proteins at ICU admission	46.7 ± 5.8	46.5 ± 5.9	0.7
Number of patients transfused, n [%]	14 [10]	16 [11]	0.8
Perioperative myocardial infarction, n [%]	1 [0.7]	3 [2]	0.6
Re-thoracotomy, n [%]	11 [7.6]	5 [3.4]	0.13
Ventilation duration (hours)	10.2 [7.2-13.3]	13.0 [7.2-18.9]	0.9

CABG: Coronary artery bypass grafting; CPB: cardiopulmonary bypass; ICU: intensive care unit.
* : surgical categories overlap and the total exceeds the number of patients in each group.

to accurately consider the true pressure contour throughout the study period. We did not bluntly force pressure to rise as a high pressure is not devoid of surgical drawbacks and, also, because some transient drops in pressure are from surgical origins and are difficult to instantly correct or fully avoid. Moreover, in some instances, the surgeon may ask for a reduced pressure. This explains why the MAP was not greater than 75 mmHg in all patients in the High Pressure group. However, blood flow was maintained constant and, thus, was likely to combine with a true average MAP of 79±6 mmHg to increase the renal perfusion regimen in the High Pressure group. Notwithstanding the higher pressure achieved in the High Pressure group, our results conflict with the those published by Rivers et al.[9] in septic shock, which showed that the early haemodynamic optimisation reduced mortality. Yeboah al.,[25] in cardiac surgery, lent support to a peroperative High Pressure regimen as they showed that a MAP below 80 mmHg, a CPB longer than 60 min and multiple valve replacement together contributed to AKI. Furthermore, in the context of cardiac surgery, Gold et al.[11] displayed promising results on the beneficial effect of an increased MAP to reduce cardiac and neurologic complications, even if they did not study renal function. All these works encouraged us to conduct the present study. However, despite achieving a 79 mmHg MAP with norepinephrine, i.e. 19 mmHg greater on average than that observed in the Control group, our results disavow such a high pressure strategy to reduce the incidence of postoperative AKI. Postoperatively, the MAP was comparable in both groups and slightly lower than expected, but matched standard pressures met in postoperative conditions. We met difficulties in achieving a sustained rise in MAP in the High Pressure group during their stay in ICU, especially because the control of this parameter could not be as stringent as during CPB. Hence, in the ICU, the MAP was not different by group, but remained in the clinical range expected in practice at this stage.

Like us, a few other studies failed to demonstrate a MAP effect. Pirraglia et al.[26] did not show any advantage of high vs. normal MAP on renal function. Urzua et al.,[12] in a randomized study including less than 20 patients, showed that maintaining the MAP above 70

Table 4. Renal function.

	Control (n=145)	High-MAP (n=147)	p value
Preoperative renal function			
Creatinine (μmol/L)	95 ± 28	93 ± 25	0.53
Indexed creatinine clearance (ml/min/1.73m²)	61 ± 21	62 ± 23	0.91
BUN (mmol/l)	8.6 ± 3.5	8.5 ± 3.3	0.85
Postoperative renal function			
Renal resistivity index	0.69 ± 0.06	0.70 ± 0.06	0.40
Renal resistivity index > 0.74 n (%)	21 (21.6)	23 (25.6)	0.53
Mean hourly diuresis (ml/kg/h)	0.95 ± 0.87	0.84 ± 0.52	0.20
Serum creatinine peak (μmolL)	108 ± 52	105 ± 50	0.73
Time of occurrence for the serum creatinine peak (h)	13.6 ± 19.1	14.2 ± 20.6	0.8
Administration of diuretics, n [%]	29 [20]	25 [17]	0.6
AKI according to 30% rise in serum creatinine, n [%]	24 [16.6]	25 [17.0]	1
AKI according to 50% rise in serum creatinine, n [%]	13 [9]	13 [8.8]	1
Classified as RIFLE “risk”, n [%]	80 [55]	92 [63]	0.2
Classified as RIFLE “injury”, n [%]	15 [10]	14 [10]	0.8
Number of patients requiring haemodialysis, n [%]	4 [2.8]	6 [4.1]	0.8
Number of dialysis/patient in ICU when required	3.0 ± 2.8	3.2 ± 1.2	0.90

BUN: blood urea nitrogen; ICU: intensive care unit; AKI: acute kidney injury; RIFLE classification: Risk, Injury, Failure, Loss, End-stage kidney disease.

mmHg resulted in a transient effect: creatinine clearance rose during CPB, but not afterwards. Similarly, Wang et al.[27] reported a positive correlation between MAP and urine flow during CPB, but it did not help in decreasing postoperative AKI. Hence, both studies do not validate raising the MAP during CPB as a means to avoid postoperative AKI.

At present, our results and the literature seem to support the inefficacy of a high pressure regimen to avoid renal injury during cardiac surgery under CPB. Moreover, it did not reduce the rate of neurological complications. Several arguments from our study seem to support this conclusion. First, we purposely selected patients at known risk for perioperative AKI. This should have, in principle, favoured the protective effect of a sustained renal perfusion pressure on renal function. Second, the CPB flow did not vary in response to the MAP as it was identical in both groups. We induced a mere MAP control, thus, allowing the assessment of the MAP effect by itself. This is worth mentioning as renal blood flow is dependent on both arterial blood pressure and CPB flow.[28] Third, we monitored renal function daily during the first five postoperative days, thus, minimizing the risks of missing any instance of renal failure, as the peak sCr increase occurs at the third postoperative day, whatever the kind of cardiac surgery performed.[5] Fourth, we tested several surrogates for renal injury (RIFLE risk and injury criteria, 30% and 50% rise in sCr). The choice of a single criterion for renal injury, especially a more stringent one, would have raised the risk of achieving a low diagnostic sensitivity. By testing several thresholds, we likely avoided such a risk. In the end, the randomization should have suppressed possible unknown bias present in previous studies.

The extracorporeal circulation duration in our study, despite inconsistently mentioned publications, appears comparable to those published[1,28–31] and was identical in both groups. Hence, this factor, which is known to influence renal integrity,[32] did not bias our results.

The rate of postoperative AKI decreased inversely to the stringency of the criteria used to define it. Comparison with the literature is difficult as the surrogates for acute renal failure vary greatly among the studies. We report a rate for AKI of about 60% when using the RIFLE risk criteria and of 3% considering the dialysis endpoint. These rates are slightly higher than those reported in large series of unselected cardiac surgery patients.[1,3,33,34] This could be explained by our deliberate selection of patients at risk for postoperative acute injury. When such risks are present, acute renal injury is known to increase in proportion to the number of risk factors the patient discloses.[33]

Before concluding, two limitations in our study should be considered. We did not achieve our goal (MAP >75 mmHg) in 18% of patients in the High Pressure group, thus, slightly reducing the true pressure difference between the groups compared to our initial goal. However, this is inherent to surgery as transient hypotension could occur, especially after surgical manipulation of the heart or at the onset or separation from CPB. However, the mean pressure difference between the groups was statistically and clinically significant. Further, raising the MAP in the High Pressure group seemed difficult to achieve and potentially dangerous. Indeed, an ultra high pressure regimen during CPB exposes the patient to the risk of bleeding and vascular tear. The goal achieved seems to correspond to what can be achieved in real life, without raising operative risks.

We chose to manipulate the MAP, but keep the cardiac output at its standard recommended level (2.4 L/min/m^2).[19] Thus, we did not test an alternate hypothesis which would have consisted of increasing cardiac output while maintaining constant MAP. Goepfert et al. showed a beneficial effect of a volume-based control of cardiac output in cardiac surgery, with most actions taking place after CPB.[35] However, they did not comment on how they handled the CPB period, during which cardiac output depends on the roller pump. To the best of our knowledge, allowing a supra-physiological "cardiac" (roller pump) output during CPB is not a validated option to preserve renal function. Moreover, raising the pump output would result in intricate changes in the MAP, possibly further altered by volume loading. We believe that such a design would be both difficult to achieve and to analyse. Nevertheless, our negative results lend support to the exploration of this field as the pressure optimization did not appear successful.

Our negative results could also result from competing factors which may have reduced the renal risk and, thus, decreased the chance of observing a difference by group. We used heparin-coated, low-volume, closed CPB circuits along with normothermia during extracorporeal circulation. Both may reduce the activation of coagulation[36] and the inflammatory cascade.[37,38] The latter is known to be detrimental to renal function[39] in addition to other unavoidable factors (ischaemia-reperfusion, emboli, the loss of blood pulsatility, etc). Moreover, we routinely processed shed blood in the operative field with a cell saver, which should allow removal by lavage of the inflammatory compounds generated in response to the air-blood contact.[40] Noteworthy, these technical "progresses" were not used in previous published studies which assessed the issue of raising the MAP during CPB.

In conclusion, our randomized, controlled study did not fully succeed in raising the MAP to the preset target value in all patients during CPB, mainly because unavoidable short drops in MAP occurred as a consequence of surgery, despite haemodynamic support by catecholamines. However, despite a significant rise in MAP in the intervention group, we did not confirm a renal protective effect of a higher MAP during the CPB period in patients with increased risk for postoperative AKI. Moreover, we routinely used inflammation-protective strategies (low volume heparin-coated CPB circuit, cell saver processing of spilled blood before reinfusion) all of which may decrease the risk of acute renal injury. Hence, MAP may become of less clinical relevance as long as such inflammation preservation strategies are warranted. This may suggest that fluid load optimisation and, perhaps, increased cardiac output may represent alternative and more convincing issues on the path of renal protection.

Declaration of Conflicting Interest

The authors declare that there is no conflict of interest.

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